# SQL ASSIGNMENT 16

**Subject:** Database Management Systems (SQL) **Topic:** Table Schemas, Relational Joins, Aggregations & Window Functions **Status:** Completed

### TASK 1

Create a table named `Restaurant` (id, name, cuisine, location, average_rating) and insert at least 5 sample rows representing popular Zomato restaurants.

```
CREATE TABLE Restaurant ( id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, cuisine VARCHAR(50) NOT NULL, location VARCHAR(100) NOT NULL, average_rating DECIMAL(3, 2) ); INSERT INTO Restaurant (id, name, cuisine, location, average_rating) VALUES (1, 'Truffles', 'Italian', 'Koramangala', 4.6), (2, 'Punjab Grill', 'North Indian', 'Indiranagar', 4.4), (3, 'Toit', 'Italian', 'Indiranagar', 4.7), (4, 'Meghana Foods', 'North Indian', 'Jayanagar', 4.5), (5, 'Empire Restaurant', 'North Indian', 'Church Street', 4.1);
```

### TASK 2

Write a SQL query to generate a report showing the count of restaurants for each cuisine type, ordered by count descending.

```
SELECT cuisine, COUNT(*) AS total_restaurants FROM Restaurant GROUP BY cuisine ORDER BY total_restaurants DESC;
```

### TASK 3

Create a `Review` table (id, restaurant_id, user_name, rating, review_date) and insert at least 10 sample reviews linked via foreign key.

```
CREATE TABLE Review ( id INT PRIMARY KEY, restaurant_id INT REFERENCES Restaurant(id), user_name VARCHAR(100), rating INT CHECK (rating BETWEEN 1 AND 5), review_date DATE ); INSERT INTO Review (id, restaurant_id, user_name, rating, review_date) VALUES (101, 1, 'Aarav', 5, '2026-10-01'), (102, 1, 'Sneha', 4, '2026-10-03'), (103, 2, 'Rohan', 4, '2026-10-05'), (104, 2, 'Priya', 5, '2026-10-06'), (105, 3, 'Ananya', 5, '2026-10-08'), (106, 3, 'Vikram', 5, '2026-10-09'), (107, 4, 'Karan', 4, '2026-10-10'), (108, 4, 'Divya', 5, '2026-10-12'), (109, 5, 'Rahul', 3, '2026-10-14'), (110, 5, 'Neha', 4, '2026-10-15');
```

### TASK 4

Write a SQL query using `JOIN` and `GROUP BY` to display each restaurant's name, cuisine, and average review rating from the `Review` table.

```
SELECT rst.name, rst.cuisine, ROUND(AVG(rev.rating), 2) AS avg_review_rating FROM Restaurant rst JOIN Review rev ON rst.id = rev.restaurant_id GROUP BY rst.id, rst.name, rst.cuisine ORDER BY avg_review_rating DESC;
```

### TASK 5

Use a window function to rank restaurants by their average review rating within each cuisine type (using `DENSE_RANK() OVER (PARTITION BY cuisine ...)`).

```
WITH RestaurantRatings AS ( SELECT rst.name, rst.cuisine, ROUND(AVG(rev.rating), 2) AS avg_rating FROM Restaurant rst JOIN Review rev ON rst.id = rev.restaurant_id GROUP BY rst.id, rst.name, rst.cuisine ) SELECT name, cuisine, avg_rating, DENSE_RANK() OVER ( PARTITION BY cuisine ORDER BY avg_rating DESC ) AS cuisine_rank FROM RestaurantRatings;
```